

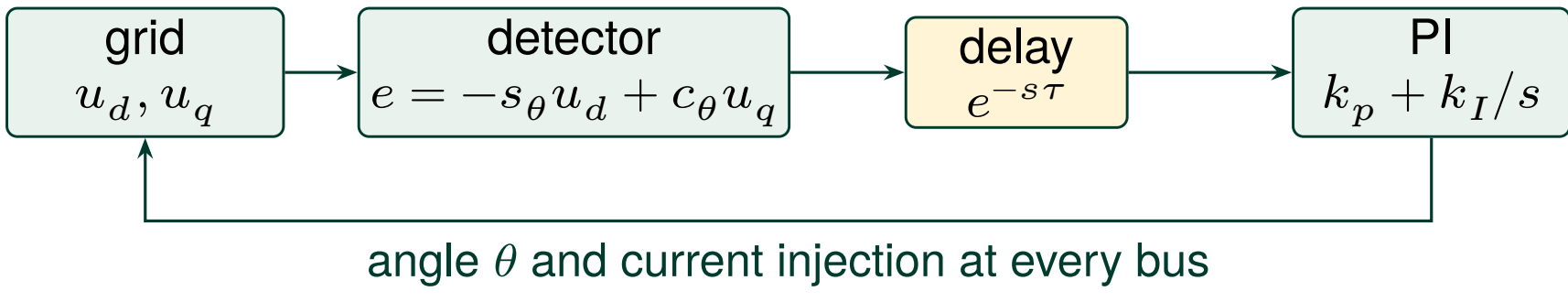
COLLECTIVE CLOSURE · EXACT DELAY TRANSPORT · PLL CO-DESIGN

BEYOND NODAL DAMPING

A local PLL change is a collective act: see the closure, then co-design the gains in the full grid
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1 THE QUESTION: A COLLECTIVE PROBLEM

Replacing a synchronous generator (SG) by a grid-following inverter (GFL) is a **local** hardware action. Its stability is a **collective** property of the network, because every PLL listens to the same grid.



Key question. Which closed loops does the network create between PLLs that are individually fine, and which gains break them?

- 1 Exact factorization: local $\times$ collective.
- 2 Damping is an operator, not a number per bus.
- 3 Exact gain synthesis and delay transport.
- 4 Co-design algorithm, tested in the full grid.

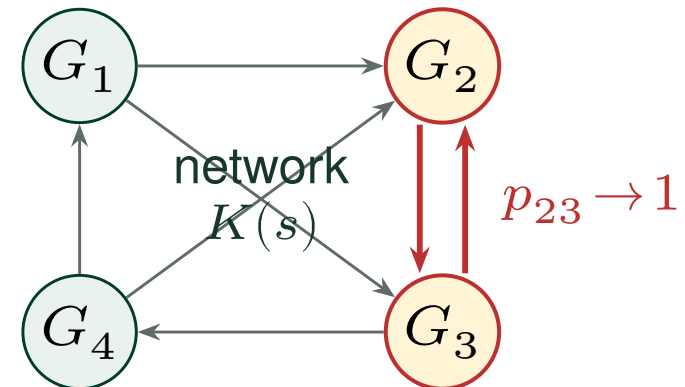
4 EXACT CLOSURE FACTORIZATION

Open the ten PLLs, keep the whole grid:

$$G = C(sI - A_0)^{-1}B, \quad L_i = 1 - G_{ii}e^{-s\tau_i}, \quad Q_{ij} = -\frac{G_{ij}e^{-s\tau_j}}{L_i}$$

$$\det \Delta = \det(sI - A_0) \prod_i L_i \cdot \det(I + Q)$$

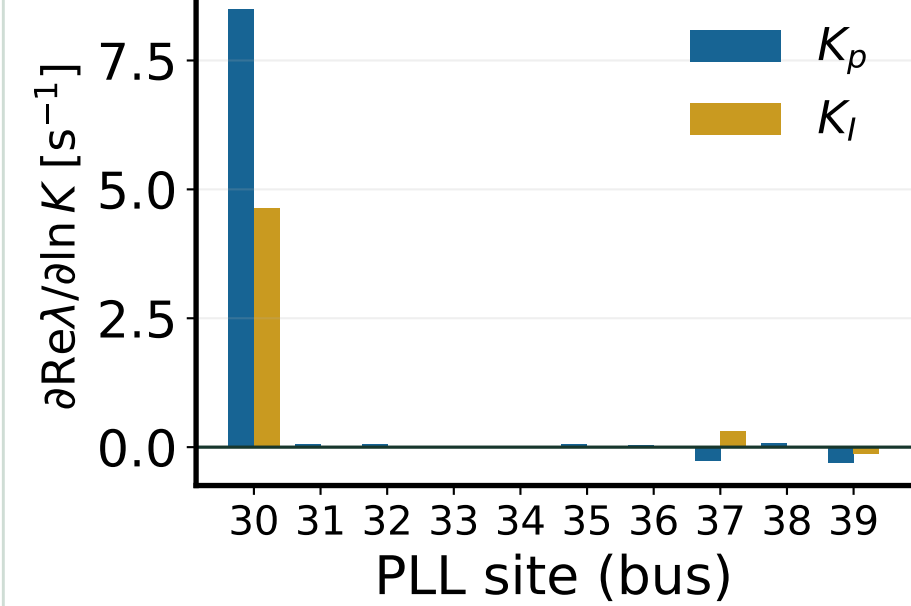
$$\det(I + Q) = 1 - \sum_{i < j} p_{ij} + \sum Q_{ij}Q_{jk}Q_{ki} - \dots, \quad p_{ij} = Q_{ij}Q_{ji}$$



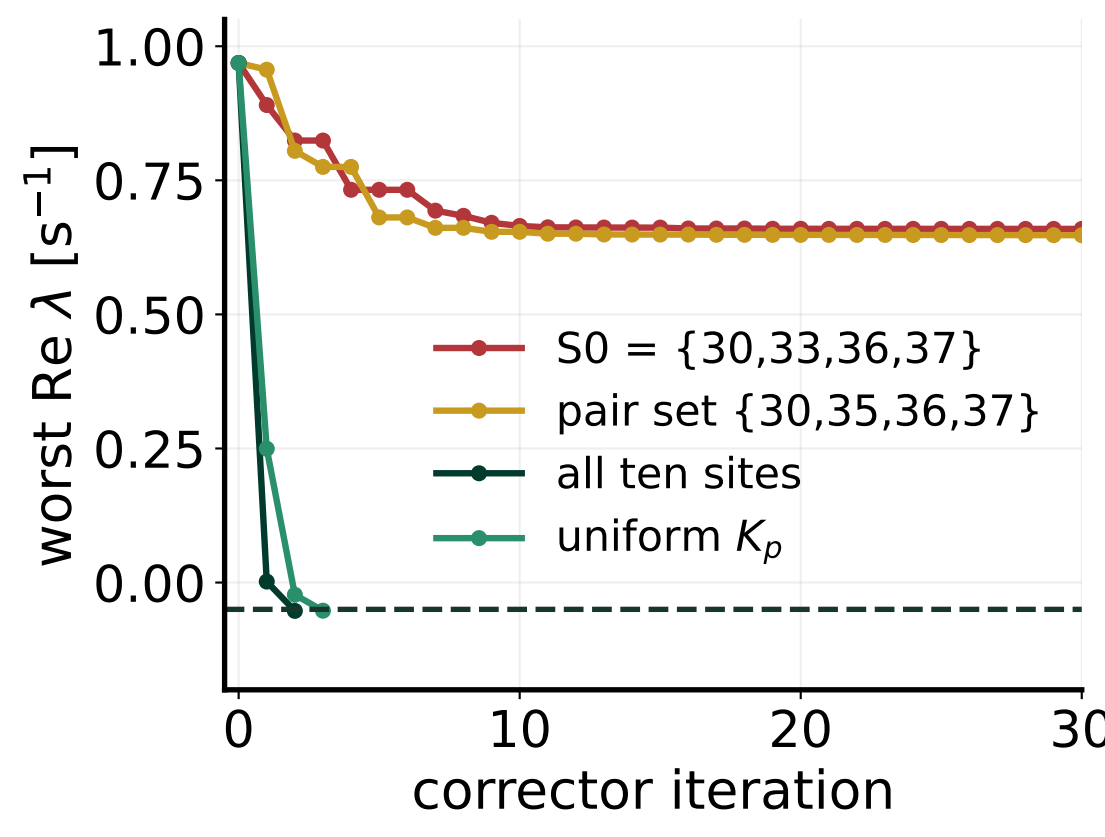
IN WORDS Delay drives $\det(I + Q)$ to zero while every L_i stays away from zero: the failure belongs to a cycle, not to a device.

7 CO-DESIGN ALGORITHM

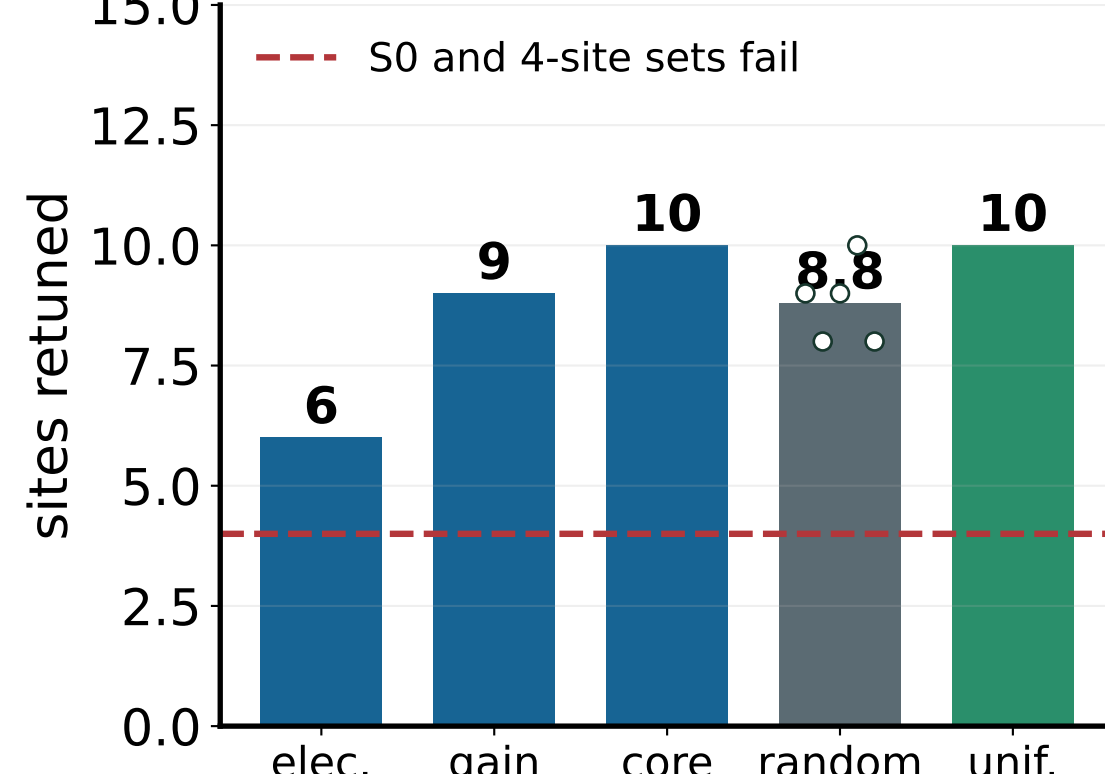
- 1 open PLLs, keep grid: $G(s)$
- 2 diagnose: eig(Q), cycles p_{ij}
- 3 gain map or transport $\lambda \mapsto (k_p, k_I)$
- 4 predict: $\min \frac{1}{2}|\delta|^2$ s.t. $\text{Re}\lambda_r + g_r \cdot \delta \leq -\sigma$
- 5 correct: exact spectrum, shrink step if worse
- 6 verify: roots + five events



Leading 44 ms root: authority sits at site 30 (8.5 s^{-1} per $\ln K_p$), not at the cycle pair 35–36 (≤ 0.05).



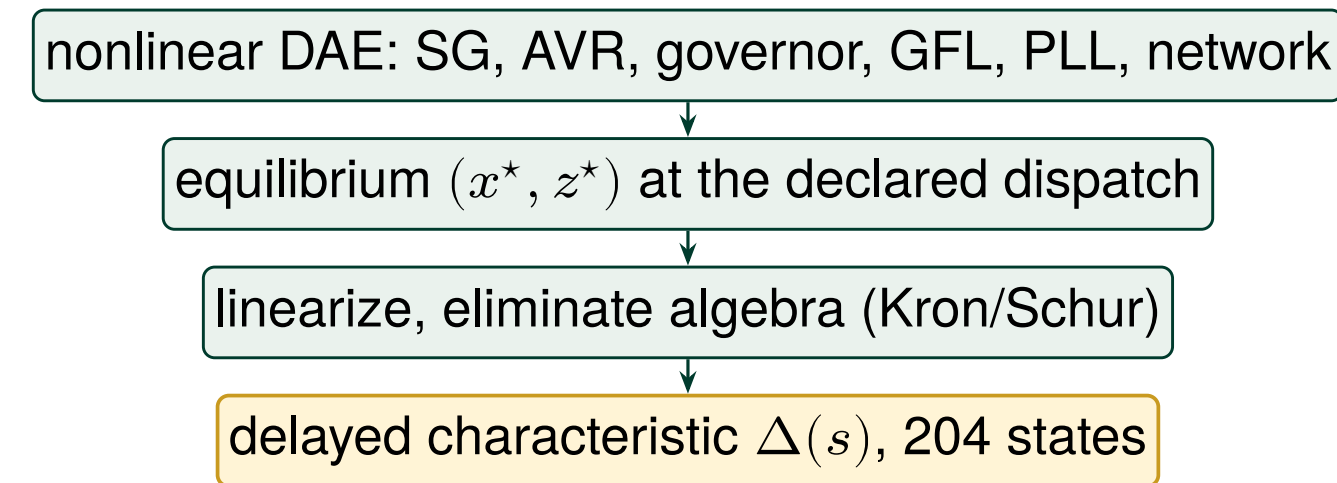
Worst root per iteration, 44 ms.



Sites retuned until the margin holds.

$S_0 = \{30, 33, 36, 37\}$ **fails** (K_I , K_p , joint); so does every four-site set. One common K_p cut of 14 % restores the margin.

2 MODEL: DAE TO DELAYED LOOP



Port operator (algebraic network seen from the retained variables):

$$T_S(s) = g_z + g_x(sI - f_x)^{-1}f_z$$

Delayed loop (PLL error is the only delayed signal):

$$\Delta(s) = sI - A_0 - B E(s) C, \quad E = \text{diag } e^{-s\tau_i}$$

A_0 : grid and devices; B : PLL injection (k_p, k_I); C : detector. Exact exponentials, never Padé. Baseline: $\rho = 0.875$, $k_p = 28.27$, $k_I = 246.7$, $\tau = 40$ ms.

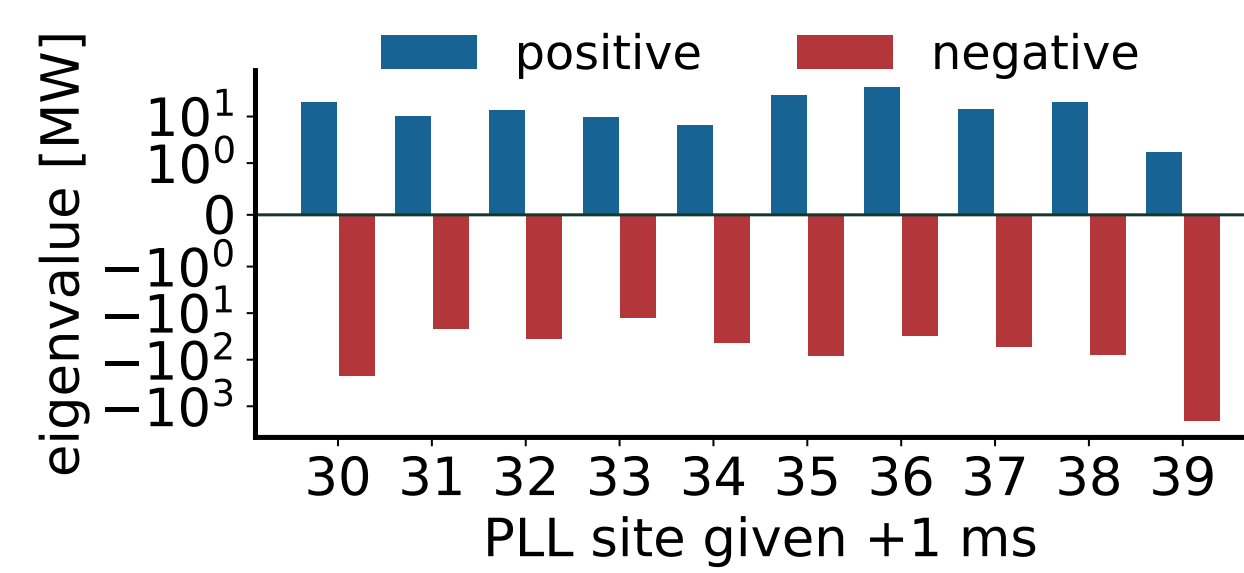
5 EFFECTIVE DAMPING OPERATORS

The controllers add a self-energy $\Pi(s)$ to the swing operator:

$$T(s) = s^2 M + s D_0 + L + \Pi(s)$$
$$M_{\text{eff}} = M + \frac{\Pi_e(s) - \Pi_e(0)}{s^2}, \quad D_{\text{eff}} = D_0 + \frac{\Pi_o(s)}{s}$$

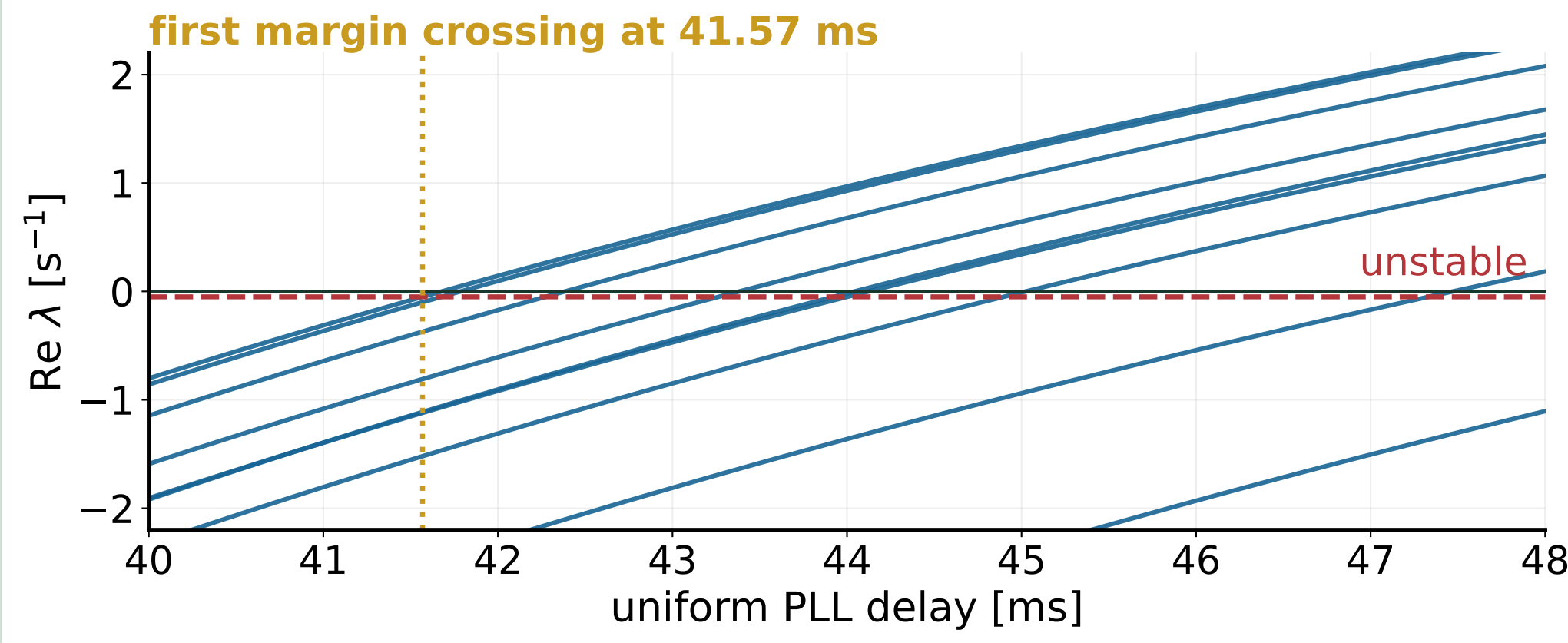
Harmonic damping supplied to the generators, and what one PLL change does to it:

$$D_H(\Omega) = \text{Herm } Z(j\Omega), \quad \delta D_H = \frac{1}{2}(a v^* + v a^*)$$

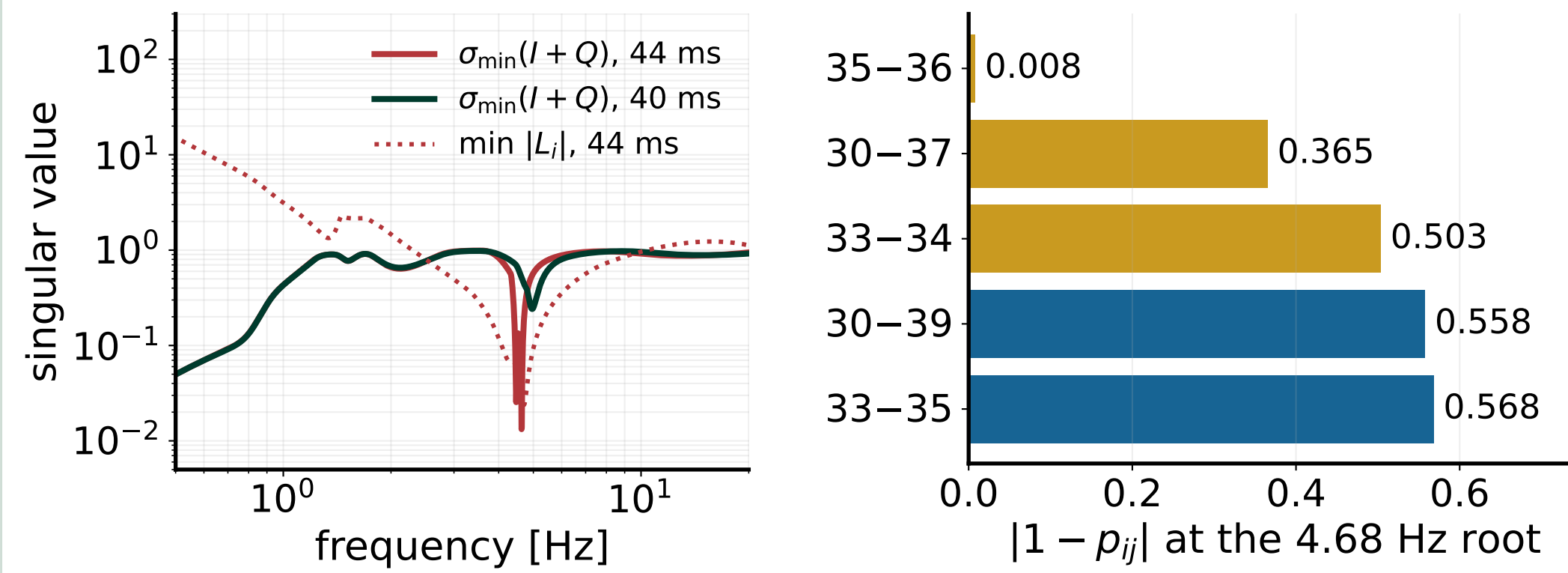


IEEE-39, +1 ms at each PLL, 5 Hz: one positive and one negative eigenvalue (60 of 60 checks).

8 FULL-GRID DELAY EVIDENCE



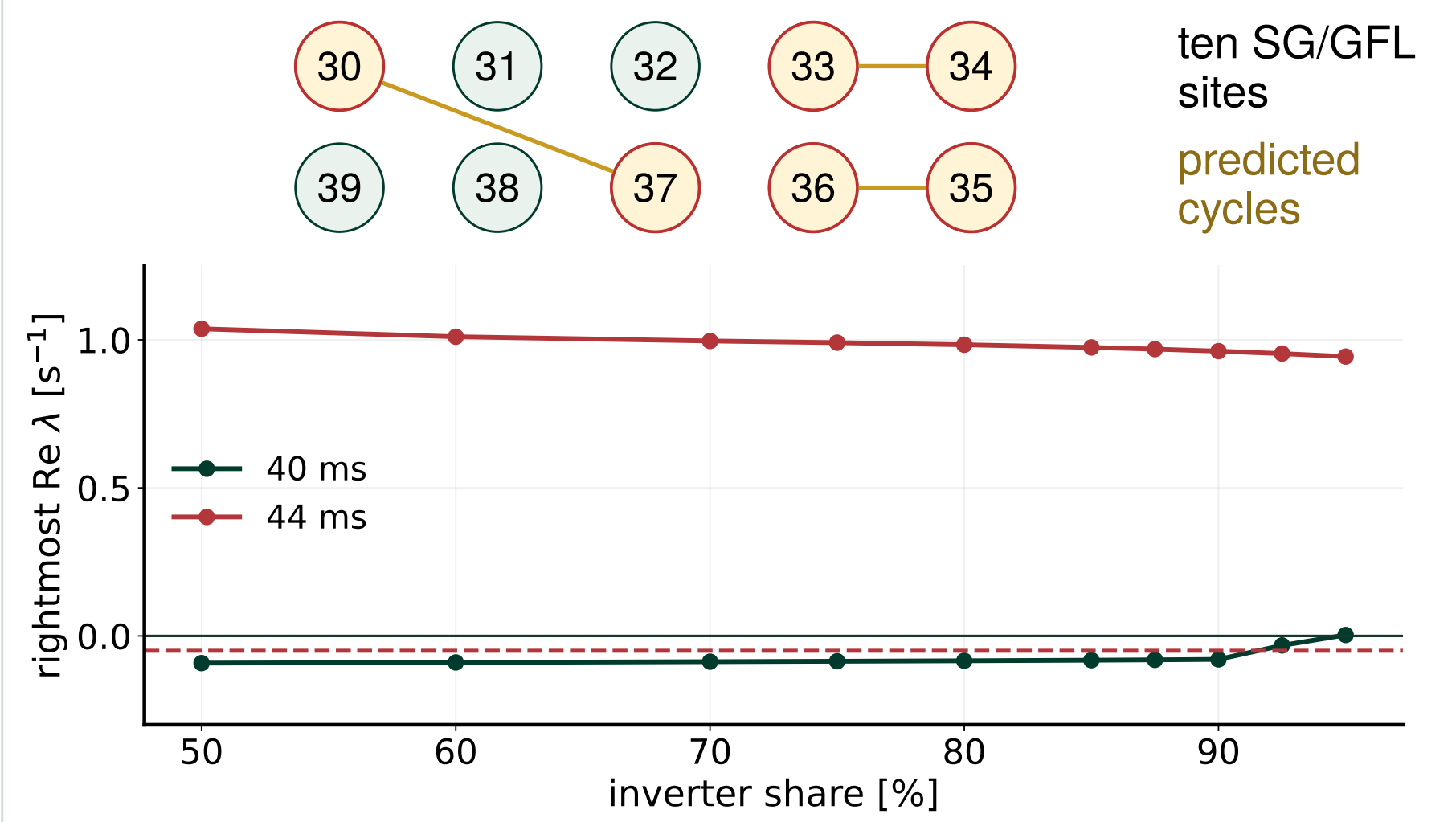
Exact-delay roots of the 4.5–5 Hz PLL family. Unstable roots: 0, 0, 4, 6, 8 at 40, 41, 42, 43, 44 ms. Absent from the all-SG grid (modes stop at 1.5 Hz).



$\sigma_{\min}(I + Q)$ dips at 4.7 Hz while local factors stay regular. Gold: predicted pairs. Closest cycles: 35–36 (0.008), 33–34, 36–37; the predicted pairs are the three strongest electrical couplings.

delay [ms] 40 41 42 43 44 46 48
unstable roots 0 0 4 6 8 14 16
beyond -0.05 0 0 4 6 10 14 16
Full-spectrum argument-principle counts (floating point, validated against eigenvalues at zero delay).

3 IEEE-39 AND INVERTER SHARE



Rightmost root versus inverter share (exact delay): at 40 ms the margin holds up to 90 % share; at 44 ms the PLL family is unstable at every share from 50 %.

6 EXACT PLL GAIN SYNTHESIS

Target mode $\lambda = \alpha + j\omega$, right vector q , $y = G(\lambda)q$:

$$W_i = \frac{\lambda^2(1 + t_f \lambda) e^{\lambda \tau_i} q_i}{y_i}, \quad k_{p,i} = \frac{\text{Im} W_i}{\omega}, \quad k_{I,i} = \text{Re} W_i - \alpha k_{p,i}$$

Delay grows by h , keep λ (exact transport):

$$k_p(h)\lambda + k_I(h) = (k_p\lambda + k_I) e^{\lambda h}$$

Price paid by every other mode μ :

$$\frac{d\mu}{dh} = -k_p(\mu - \lambda)(\mu - \bar{\lambda}) \frac{\partial \mu}{\partial k_I}, \quad \frac{\partial \mu}{\partial k_I} = -\frac{\ell^H Z_{k_I} r}{\ell^H Z_s r}$$

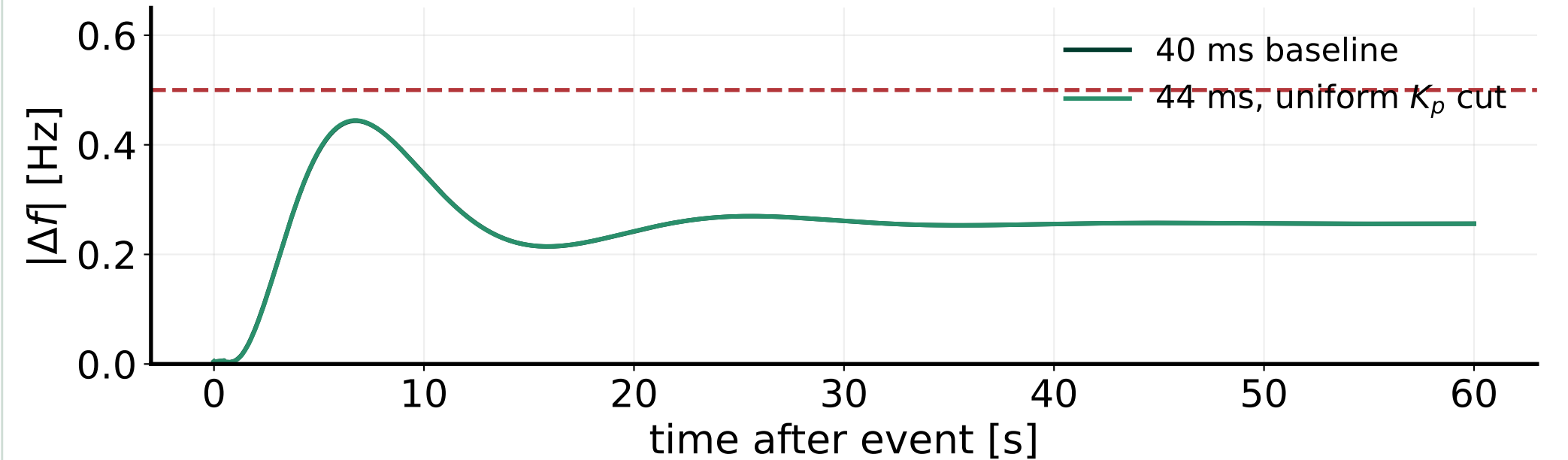
Rule (fixed in advance): among modes predicted safe, protect the one that keeps the most k_I . 9-bus bank: $+0.886 \rightarrow -0.131 \text{ s}^{-1}$.

IN WORDS PLL: quadratic factor. Grid: residue.

9 NONLINEAR CHECKS AND LIMITS



Forced response, 5 MW at bus 29, $f^* = 4.65$ Hz (Julia, exact delay).



Load step +100 MW at bus 16: max bus-frequency deviation.

design	events	max $ \Delta f $	slack
baseline, 40 ms	5/5	0.471	0.0074
uniform K_p -14 %, 44 ms	5/5	0.472	0.0072
6-site sparse, 44 ms	4/5	0.492	0.0014
no retune, 44 ms	abort	—	—

Limits. Floating-point counts, not certified; four-site retuning and the reduced-model 7–8 Hz family were refuted; five events are a finite campaign.

TAKE-HOME The closure is local (cycle 35–36, strongest couplings); the repair is global (one common K_p cut). Sparse retuning of the core did not work in the full grid.

TAKEAWAY

PRESERVING ONE MODE
DOES NOT PROTECT THE GRID.

The gains that hide a delay from one mode are exact. The network decides what they do to all the others.

REFERENCES

- [1] Marković et al., IEEE Trans. Power Syst., 2021.
- [2] Huang et al., arXiv:1903.05489.
- [3] Michiels & Gumussoy, arXiv:2003.05496.
- [4] Dörfler et al., IEEE Trans. Power Syst., 2014.
- [5] Bindel & Hood, SIAM J. Matrix Anal. Appl., 2013.
- [6] Johansson, IEEE Trans. Comput., 2017.

Topics: [1] converter small-signal stability; [2] PLL stability and grid structure; [3] fixed-order design with delays; [4] sparse wide-area control; [5] nonlinear eigenvalues; [6] interval arithmetic.

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